

main diagonal, 0 on the main diagonal except the upper right corner, and corresponding to a zero-triangle graph in the upper right corner.

Limits of such graphons in the metric must take the form

$$(15) \quad g(x, y) = \begin{cases} 1 & x < kc < y \text{ or } y < kc < x \text{ for an integer } k < \ell; \\ 0 & (k-1)c < x, y < kc \text{ for some integer } k < \ell; \\ \text{unspecified} & x, y > (\ell-1)c, \end{cases}$$

with

$$(16) \quad \iiint_{[(\ell-1)c, 1]^3} g(x, y)g(y, z)g(z, x) dx dy dz = 0,$$

and with $\iint_{[0, 1]^2} g(x, y) dx dy = e$. Minimizing $I(g)$ on such graphons is easy, since all but the upper right corner of the graphon is fixed. Applying Theorem 4.1 to that corner, we get

Theorem 4.2. *If $e > 1/2$ and t is the smallest value possible, then the minimum of $I(g)$ on F_0 is achieved by the graphon*

$$(17) \quad g_0(x, y) = \begin{cases} 1 & x < kc < y \text{ or } y < kc < x \text{ for an integer } k < \ell; \\ p & (\ell-1)c < x < [1 + (\ell-1)c]/2 < y \text{ or } (\ell-1)c < y < [1 + (\ell-1)c]/2 < x; \\ 0 & \text{otherwise,} \end{cases}$$

where

$$(18) \quad p = \frac{4c(1 - \ell c)}{(1 - (\ell-1)c)^2}$$

is a number chosen to make $\iint_{[0, 1]^2} g(x, y) dx dy = e$. Furthermore, any other minimizer is equivalent to g_0 . The minimum value of $I(g)$ is

$$(19) \quad I(g_0) = \frac{(1 - (\ell-1)c)^2}{2} I_0(p).$$

5. MINIMIZING NEAR THE FIRST SCALLOP

Now that we know the minimizer at the (lower) boundary, we perturb it to get a minimizer near the boundary.

Theorem 5.1. *Pick $e < 1/2$ and ϵ sufficiently small. Then the graphon*

$$(20) \quad g(x, y) = \begin{cases} 2e - \epsilon & x < \frac{1}{2} < y \text{ or } y < \frac{1}{2} < x \\ \epsilon & \text{otherwise,} \end{cases}$$

minimizes the rate function to second order in perturbation theory among graphons with $e(g) = e$ and $t(g) = e^3 - (e - \epsilon)^3$. For pointwise small variations δg of g , the second variation in $I(g)$ is bounded from below by $\frac{1}{2} \iint_{[0, 1]^2} (\delta g(x, y))^2 dx dy$.

Proof. We first consider the first variation in $I(g)$ for general graphons and derive the Euler-Lagrange equations. It is easy to check that our specific g satisfies these equations. We then consider the second variation in $I(g)$. Note that the function I_0 satisfies

$$(21) \quad I'_0(u) = \frac{1}{2}[\ln(u) - \ln(1-u)], \quad I''_0(u) = \frac{1}{2} \left[\frac{1}{u} + \frac{1}{1-u} \right] \geq 2.$$

To find the Euler-Lagrange equations with the constraints that $(e(g), t(g))$ are equal to fixed values (e_0, t_0) , we use Lagrange multipliers and vary the function $I(g) + \lambda_1(e(g) - e_0) + \lambda_2(t(g) - t_0)$. To first order, the variation with respect to g is

$$(22) \quad \delta I(g) = \int_0^1 \int_0^1 I'_0(g(x, y)) \delta g(x, y) dx dy + \lambda_1 \int_0^1 \int_0^1 \delta g(x, y) dx dy$$

$$(23) \quad + 3\lambda_2 \int_0^1 \int_0^1 h(x, y) \delta g(x, y) dx dy,$$

where we have introduced the auxiliary function

$$(24) \quad h(x, y) = \int_0^1 g(x, z)g(y, z) dz.$$

Setting $\delta I(g)$ equal to zero, we get

$$(25) \quad I'_0(g(x, y)) = -\lambda_1 - 3\lambda_2 h(x, y).$$

Our particular $g(x, y)$ satisfies this equation with

$$(26) \quad 3\lambda_2 = \frac{I'_0(2e - \epsilon) - I'_0(\epsilon)}{2(e - \epsilon)^2}.$$

Next we expand δt and δI to second order in δg , ignoring $O((\delta g)^3)$ terms. Since

$$(27) \quad \delta t = 3 \iint h(x, y) \delta g(x, y) dx dy + 3 \iiint g(x, y) \delta g(x, z) \delta g(y, z) dx dy dz + O((\delta g)^3),$$

and since we are holding $e(g)$ and $t(g)$ fixed,

$$(28) \quad \begin{aligned} \delta I &= \iint I'_0(g(x, y)) \delta g(x, y) dx dy \\ &\quad + \frac{1}{2} \iint I''_0(g(x, y)) (\delta g(x, y))^2 dx dy \\ &= \iint (-\lambda_1 - 3\lambda_2 h(x, y)) \delta g(x, y) dx dy \\ &\quad + \frac{1}{2} \iint I''_0(g(x, y)) (\delta g(x, y))^2 dx dy \\ &= -\lambda_1 \delta e - \lambda_2 \delta t + 3\lambda_2 \iiint g(x, y) \delta g(x, z) \delta g(y, z) dx dy dz \\ &\quad + \frac{1}{2} \iint I''_0(g(x, y)) (\delta g(x, y))^2 dx dy \\ &= 3\lambda_2 \iiint g(x, y) \delta g(x, z) \delta g(y, z) dx dy dz \\ &\quad + \frac{1}{4} \iint I''_0(g(x, y)) \delta g(x, y)^2 dx dy + \frac{1}{4} \iint I''_0(g(x, y)) \delta g(x, y)^2 dx dy. \end{aligned}$$

We have split the $\iint I'' \delta g^2$ term into two pieces, as we will be applying different estimates to each piece.

Since $h(x, y)$ and $I''(g)$ are piecewise constant, all of our integrals break down into integrals over different quadrants. Let R_1 and R_2 be the following subsets of $[0, 1]^2$:

$$(29) \quad R_1 = \{x, y < 1/2\} \cup \{x, y > 1/2\}, \quad R_2 = \{x < 1/2 < y\} \cup \{y < 1/2 < x\}.$$

For each z , we define the functions $f_1(z) = \int_0^{1/2} \delta g(x, z) dx$ and $f_2(z) = \int_{1/2}^1 \delta g(x, z) dx$. The second variation in I is then

$$\begin{aligned} & \frac{1}{4} \iint_{[0,1]^2} I''_0(g) \delta g(x, y)^2 dx dy + \frac{I''_0(\epsilon)}{4} \iint_{R_1} \delta g(x, y)^2 dx dy + \frac{I''_0(2e - \epsilon)}{4} \iint_{R_2} \delta g(x, y)^2 dx dy \\ & + 3\lambda_2 \int_0^1 dz \left[\epsilon \iint_{R_1} \delta g(x, z) \delta g(y, z) dx dy + (2e - \epsilon) \iint_{R_2} \delta g(x, z) \delta g(y, z) dx dy \right] \\ & = \frac{1}{4} \iint_{[0,1]^2} I''_0(g(x, y)) \delta g(x, y)^2 dx dy + \frac{I''_0(\epsilon)}{4} \iint_{R_1} \delta g(x, z)^2 dx dz + \frac{I''_0(2e - \epsilon)}{4} \iint_{R_2} \delta g(x, z)^2 dx dz \\ (30) \quad & + 3\lambda_2 \int_0^1 \epsilon [f_1(z)^2 + f_2(z)^2] + 2(2e - \epsilon) f_1(z) f_2(z) dz \end{aligned}$$

Note that by Cauchy-Schwarz,

$$(31) \quad \int_0^{1/2} (\delta g(x, z))^2 dx \geq 2 \left(\int_0^{1/2} \delta g(x, z) dx \right)^2 = 2f_1(z)^2$$

$$(32) \quad \int_{1/2}^1 (\delta g(x, z))^2 dx \geq 2 \left(\int_{1/2}^1 \delta g(x, z) dx \right)^2 = 2f_2(z)^2.$$

Since $I''_0(\epsilon)$ and $I''_0(2e - \epsilon)$ are positive, δI is bounded from below by

$$\begin{aligned} & \frac{1}{4} \iint_{[0,1]^2} I''_0(g(x, y)) \delta g(x, y)^2 dx dy + \frac{I''_0(\epsilon)}{2} \left[\int_0^{1/2} f_1(z)^2 dz + \int_{1/2}^1 f_2(z)^2 dz \right] \\ & + \frac{I''_0(2e - \epsilon)}{2} \left[\int_0^{1/2} f_2(z)^2 dz + \int_{1/2}^1 f_1(z)^2 dz \right] \\ (33) \quad & + 3\lambda_2 \int_0^1 dz [\epsilon (f_1(z)^2 + f_2(z)^2) + 2(2e - \epsilon) f_1(z) f_2(z)] \end{aligned}$$

Collecting terms and applying equation (26), this bound becomes

$$\begin{aligned} & \frac{1}{4} \iint_{[0,1]^2} I''_0(g(x, y)) \delta g(x, y)^2 dx dy + \int_0^{1/2} dz [c_1 f_1(z)^2 + c_2 f_2(z)^2 + 2c_3 f_1(z) f_2(z)] \\ (34) \quad & + \int_{1/2}^1 dz [c_1 f_2(z)^2 + c_2 f_1(z)^2 + 2c_3 f_1(z) f_2(z)], \end{aligned}$$

where

$$(35) \quad c_1 = \frac{I_0''(\epsilon)}{2} + \frac{\epsilon(I_0'(2e - \epsilon) - I_0'(\epsilon))}{2(e - \epsilon)^2}$$

$$(36) \quad c_2 = \frac{I_0''(2e - \epsilon)}{2} + \frac{\epsilon(I_0'(2e - \epsilon) - I_0'(\epsilon))}{2(e - \epsilon)^2}$$

$$(37) \quad c_3 = \frac{(2e - \epsilon)(I_0'(2e - \epsilon) - I_0'(\epsilon))}{2(e - \epsilon)^2}.$$

Note that all coefficients are positive, and that $c_2 > 1$. As $\epsilon \rightarrow 0$, c_1 goes to $+\infty$ as $1/\epsilon$, while c_3 only diverges as $-\ln(\epsilon)$. Since $c_1 c_2 > c_3^2$ for small ϵ , the integrand for each z is positive semi-definite, so the integral over z is non-negative, and we obtain

$$(38) \quad \delta I \geq \frac{1}{4} \iint I_0''(g) \delta g^2 \geq \frac{1}{2} \iint \delta g(x, y)^2,$$

where we used the fact that $I_0''(u) \geq 2$ for all u . $\square$

Any global minimizer must be $O(\epsilon)$ close to g_0 , and hence $O(\epsilon)$ close to our specified perturbative minimizer. This means that the only way for them to differ is through a complicated bifurcation of minimizers at g_0 , despite the uniform bounds on δI as we approach the boundary. The difference between these hypothetical minimizers and g_0 would not be pointwise small, but would merely be small in an L^1 sense.

For instance, consider graphons of the form

$$(39) \quad g(x, y) = \begin{cases} p & x < c < y \text{ or } y < c < x; \\ \alpha & x, y < c; \\ \beta & x, y > c, \end{cases}$$

where c is a parameter that we will vary and p , α and β are constants that depend on c . For each c sufficiently close to $1/2$, it is possible to find a graphon of this form such that $\iiint g(x, y)g(y, z)g(x, z)dx dy dz = t$ and $\iint g(x, y)dx dy = e$, and such that the Euler-Lagrange equations (25) are satisfied. Call this graphon $g_c(x, y)$. A lengthy calculation shows that

$$(40) \quad \left. \frac{\partial^2 I(g_c)}{\partial c^2} \right|_{c=1/2} \geq 16e^2$$

for small t , indicating that (nearly) bipartite graphs with pieces of unequal size have a higher rate function than g . This provides strong evidence that our perturbative solution is in fact a global minimizer for sufficiently small t .

Corollary 5.2. *Assuming our perturbative solution is the global optimizer, there is a phase transition near the boundary point $(1/2, 0)$ between the first and second scallop.*

Proof. Our perturbative solution yields a formula for the entropy:

$$(41) \quad s_{e,t} = -\frac{1}{2}[I_0(\epsilon) + I_0(2e - \epsilon)].$$

This formula for the entropy cannot be extended analytically beyond $e = (1 + \epsilon)/2$, as $\partial^2 s / \partial e^2$ diverges as $e \rightarrow (1 + \epsilon)/2$. However, $e = (1 + \epsilon)/2$ corresponds to $t = (\epsilon^3 + 3\epsilon)/4$, or, using the more basic variable e ,

$$(42) \quad t = [(2e - 1)^3 + 3(2e - 1)]/4,$$

which is in the interior of (e, t) space. (Since the graphon $g(x, y)$ is nowhere zero, it differs in form from the graphons describing graphs with minimal t .) Thus $s_{e,t}$ must fail to be analytic in some neighborhood of the first scallop. $\square$

Of course there must also be a phase transition, presumably from this bipartite phase to a homogeneous phase, if one fixes e and raises t , which we see as follows.

Corollary 5.3. *Assuming our perturbative solution is the global optimizer, there is a phase transition as one raises t , for any fixed $0 < e < 1/2$.*

Proof. Recall from Corollary 5.2 the connection between t and ϵ :

$$(43) \quad t = \epsilon^3 - (e - \epsilon)^3.$$

Note that t is an increasing function of ϵ and reaches the value $2e^3$ when $\epsilon = 2e$. From equation (41) for the entropy we see that it cannot be extended analytically to $t > 2e^3$, yet for $e < 1/2$ we have $2e^3 < e^2 < e^{3/2}$ so (e, t) is in the interior of the phase space (Figure 1). $\square$

6. CONCLUSION

Our goal was to analyze possible phase transitions between multipartite phases of complex networks, analogous to solid/solid transitions in materials. To this end we adapted the Strauss model [St], defined in the grand canonical ensemble, to a microcanonical ensemble. It is appropriate at this point to review the ‘equivalence of ensembles’ in statistical physics.

In thermodynamics the concavity of the entropy $S(E, N, V)$, as a function of internal energy E , particle number (or mass) N and volume V , and the interpretation of equilibrium states as states maximizing the entropy, are both fundamental; see for instance [Ca, Ma]. Lagrange multipliers can be used to convert this optimization criterion of the entropy to an equivalent optimization criterion of the free energy, where the free energy is the Legendre transform of the entropy [Do, Ca, Ma]. It is important that the Legendre transform between the entropy and free energy be invertible so the two optimization schemes are equivalent, and this follows from the concavity of the entropy. (See Section 26 of [Ro] for the mathematics of the Legendre transform between convex functions.)

Statistical mechanics supplies a model for thermodynamic states, as probability distributions on mechanical multiparticle states. From a given short range particle interaction one can then (in principle) compute the internal energy E and entropy S , and prove the above two features of the entropy: its concavity and its optimization role for equilibrium states. To do this one uses the basic Boltzmann/Gibbs ansatz: that the entropy $S(E, N, V)$ is proportional to

$$(44) \quad - \sum_j \rho_j \ln(\rho_j)$$